P1

(a) True.

when the original problem has infinite optimal solutions.

$$\begin{aligned} &\text{minimize} && x_1+x_2 \\ &\text{s.t} && x_1+x_2=1 \\ & && x_1, x_2 \geq 0 \end{aligned} \quad \longrightarrow \quad \begin{aligned} &\text{minimize} && y \\ &\text{s.t.} && x_1+x_2+y=1 \\ & && x_1, x_2, y \geq 0 \end{aligned}$$

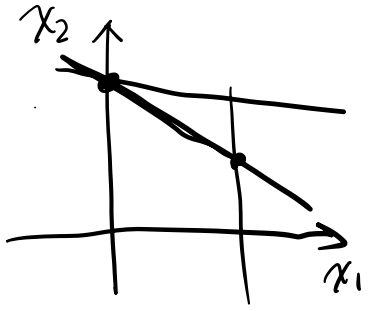
$m=1$, (x_1, x_2, y) can be $(0.5, 0.5, 0)$. 2 non-zeros $> m$

(b) False. phase I is only set to test the feasibility of the problem and get an initial BFS.

the unboundedness happenes in phase II, if the incoming column j $\forall i=1,\dots,m$. $A_{ij} \leq 0$, then the optimal value is $-\infty$.

(c) False

$$\begin{aligned} &\text{minimize} && -(x_1+2x_2) \\ &\text{s.t.} && x_1+x_2+s_1=2 \\ & && x_1+s_2=0 \\ & && x_1+s_3=1 \\ & && x_2+s_4=0 \end{aligned} \qquad m=4$$



$$x^* = (x_1, x_2, s_1, s_2, s_3, s_4)$$

$$= (0, 2, 0, 0, 1, -2)$$

there is no other optimal solution

(d) True

if a basic solution is feasible, then it is a BFS, then it must be an extreme point in the set $\{x : Ax=b,\ x \geq 0\}$. so contradict!

Thus, when a BS is not an extreme point of the set $\{x : Ax=b,\ x \geq 0\}$. it is not feasible

P2. minimize $[-1,-2,-3,-4]\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

s.t. $\begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 4 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$x \geq 0$

rank(A) = 3,

B	(-1)	-2	-3	-4	0	0	0	0
5	(1)	1	1	0	(1)	0	0	1
6	(0)	0	1	-1	(0)	1	0	2
7	(0)	1	-2	4	(0)	0	1	3

$x_B = [0\ 0\ 0\ 1\ 2\ 3]^T$

↓

B	0	(-1)	-2	-4	1	0	0	1
1	(1)	(1)	1	0	1	0	0	1
6	(0)	(0)	1	-1	0	1	0	2
7	(0)	(1)	-2	4	0	0	1	3

$x_B = [1\ 0\ 0\ 0\ 2\ 3]^T$

↓

B	1	0	−1	−4	2	0	0	2
2	1	1	1	0	1	0	0	1
6	0	0	1	−1	0	1	0	2
7	−1	0	−3	4	−1	0	1	2

$x_B = [0\ 1\ 0\ 0\ 0\ 2\ 3]^T$

↓

B	2	1	0	−4	3	0	0	3
3	1	1	1	0	1	0	0	1
6	−1	−1	0	−1	−1	1	0	1
7	2	3	0	4	2	0	1	5

$x_B = [0\ 0\ 1\ 0\ 0\ 2\ 3]^T$

↓

B	4	4	0	0	5	0	0	8
3	1	1	1	0	1	0	0	1
6	−1/2	−1/4	0	0	−1/2	1	1/4	9/4
4	2	3	0	4	2	0	1	5

$x_B = [0\ 0\ 1\ \tfrac{5}{4}\ 0\ \tfrac{9}{4}\ 0]^T$

So $x = [0, 0, 1, \frac{5}{4}, 0, \frac{9}{4}, 0]^T$

optimal value is −8

P_3 the initial problem can be referred as

B	1	1	0	2	0	0	0
	1	-1	0	0	-1	0	1
	1	1	-1	-1	0	1	2

↓

minimize $s_2 + s_4$

s.t. $x_1 - x_2 - s_1 + s_3 = 1$

$x_1 + x_2 - x_3 - x_4 + s_2 + s_4 = 2$

$x_1, x_2, x_3, x_4, s_1, s_2, s_3, s_4 \geq 0$

phase I

B	-2	0	1	1	1	-1	0	0	-3
7	1	-1	0	0	-1	0	1	0	1
8	1	1	-1	-1	0	1	0	1	2

B	0	-2	1	1	-1	-1	2	0	-1
1	1	-1	0	0	-1	0	1	0	1
8	0	2	-1	-1	1	1	-1	1	1

B	0	0	0	0	0	0	1	1	0
1	1	0	-1/2	-1/2	-1/2	1/2	1/2	1/2	3/2
2	0	2	-1	-1	1	1	-1	1	1

the original problem is feasible

$x = (3/2, 1/2, 0, 0, 0, 0, 0, 0)$

phase II

$$\tilde{C}^T = C^T - C_B^T A_B^{-1} A = [1\ 1\ 0\ 2\ 0\ 0] - [1/2, 1/2]\, A$$

$$= [1\ 1\ 0\ 2\ 0\ 0] - [1, 0, -1/2, -1/2, -1/2, 1/2]$$

$$= [0, 1, 1/2, 5/2, 1/2, 1/2]$$

B	0	1	1/2	5/2	1/2	1/2	-2
1	1	-1	0	0	-1	0	1
2	1	1	-1	-1	0	1	2

Thus, $x = (3/2, 1/2, 0, 0, 0, 0)$

optimal value is 2

P4

minimize $x_2 - 4x_3 + x_5$

s.t $x_1 + x_2 + x_3 + x_4 = 1$

$-(x_2 + x_3 + x_4 - x_5) = 1$

$x_1 + 2x_3 - x_5 - s_1 = 0$

$(x_1, \ldots, x_5, s_1) \geq 0$

phase I

minimize $s_2 + s_3 + s_4$

s.t $x_1 + x_2 + x_3 + x_4 + s_2 = 1$

$-x_2 - x_3 - x_4 + x_5 + s_3 = 1$

$x_1 + 2x_3 - x_5 - s_1 + s_4 = 0$

$(x_1 \ldots x_5, s_1 \ldots s_4) \geq 0$

B	-2	-2	0	0	0	1	0	0	0	-2
7	1	1	1	1	0	0	1	0	0	1
8	0	-1	-1	-1	1	0	0	1	0	1
9	1	2	0	0	-1	-1	0	0	1	0

↓

B	0	0	2	2	0	0	2	0	0	0
1	1	1	1	1	0	0	1	0	0	1
8	0	-1	-1	-1	1	0	0	1	0	1
9	0	-1	1	1	1	1	-1	0	-1	1

↓

B	0	0	2	2	0	0	2	0	0	0
1	1	1	1	1	0	0	1	0	0	1
5	0	-1	-1	-1	1	0	0	1	0	1
6	0	0	2	2	0	1	-1	-1	-1	0

$x_1 = 1$

$x_5 = 1$

$s_1 = 0$

phase 2:

$$\bar{c}^T = c^T - c_B^T A_B^{-1} A = [0\ 1\ -4\ 0\ 1\ 0] - [0\ 1\ 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} A$$

$$= [0\ 1\ -4\ 0\ 1\ 0] - [0, -1, -1, -1, 1, 0]$$

$$= [0, 2, -3, 1\ 0\ 0]$$

B	0	2	-3	1	0	0	-1
1	1	1	1	1	0	0	1
5	0	-1	-1	-1	1	0	1
6	1	0	2	0	-1	-1	0

B	3	5	0	4	0	0	2
3	1	1	1	1	0	0	1
5	1	0	0	0	1	0	2
6	1	2	0	2	1	1	2

$x_3 = 1$

$x_5 = 2$ so $x = [0\ 0\ 1\ 0\ 2\ 0]$

$s_1 = 0$ optimal value is -2

P5

B	0	1	α	0	0	β	-1	0
4	0	1	0	1	0	1	0	η
5	0	-1	-1	0	1	0	δ	5
1	1	0	γ	0	0	-1	0	0

(a) there is a -1 on upper left.

so it is not optimal

$x_1 = 0$, $x_4 = \eta$, $x_5 = 5$

$\therefore \eta \geq 0$, the BS is feasible

(b) when $A_{ij} \leq 0$ to every ith row in the incoming column, the problem is unbounded.

① when col 3 is the incoming column,
$\alpha < 0$, $\gamma \leq 0$, the problem is unbounded

② when col 6 is the incoming column,
$\alpha \geq 0$, $\beta < 0$, $\eta < 0$

③ when col 7 is the incoming column,
$\alpha \geq 0$, $\beta \geq 0$, $\delta < 0$

(c) $\alpha \geq 0$, $\beta = -1$, $\eta = 1$

or $\alpha \geq 0$, $\beta \geq 0$, $\delta = 5$

(d) ① $\alpha \geq 1$, $\beta \geq 0$, $\delta > 0$

② $\alpha < 0$, $\gamma \leq 0$

② $\alpha \geq 0$, $\beta < 0$, $\eta < 0$

③ $\alpha \geq 0$, $\beta \geq 0$, $\delta < 0$